Fiscal Year:	FY 2014	**Task Last Updated:**	FY 06/02/2014
PI Name:	Baulch, Janet E. Ph.D.		
Project Title:	High LET Radiation Induced Carcinogenesis: Epigenetic Mechanisms in Mouse Models of Human Cancers		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jbaulch@uci.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	949.824.7396
Organization Name:	University of California, Irvine		
PI Address 1:	Department of Radiation Oncology		
PI Address 2:	Medical Sciences I, B149		
PI Web Page:			
City:	Irvine	**State:**	CA
Zip Code:	92697	**Congressional District:**	45
Comments:	NOTE: PI previously at University of Maryland; moved to UCA Irvine in summer 2013 (Ed., 11/11/13)		
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Radiobiology NNJ11ZSA001N
Start Date:	08/01/2013	**End Date:**	07/31/2016
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 7/31/2016 per NSSC information (Ed., 7/14/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AK69G		
Performance Goal No.:			
Performance Goal Text:			
	[Editor's note 11/11/2013. Note this is a continuation of "High LET Radiation Induced Carcinogenesis: Epigenetic Mechanisms in Mouse Models of Human Cancers" (grant NNX12AE79G); see that project for previous reports.] This is a collaborative project with other NASA investigators who are characterizing mouse models of human lung and colon cancers. Data from both our laboratory and the literature suggest that epigenetics play a significant role in carcinogenic processes. By describing the changes in DNA methylation and miR regulation of gene expression for these mice, we will provide NASA with mechanistic data regarding carcinogenic initiation and progression. By using normal, cancer, and metastatic cancer cell lines we will obtain information regarding epigenetic changes that occur during carcinogenesis based on stage. For example, loss of control of gene expression for extracellular matrix (ECM) remodeling genes may foster metastatic behavior in colon cancer. For this reason we will include ECM gene promoter		

Task Description:	DNA methylation and gene expression among the loci evaluated in this study to the hypothesis that dis-regulation of epigenetic marks and gene expression changes with disease progression. Similarly, the work of other NASA investigators suggests that we will see epigenetically regulated differences among cell lines in expression of inflammatory response pathway gene regulation among lung cell lines. These experiments will provide information regarding radiation responses of these mouse tissues and human cell lines based on the LET of the radiation used. Abnormal epigenetic silencing of genes can occur at a variety of times during tumor progression. If so, does radiation exposure push normal cells to a cancer phenotype and cancer cells to a metastatic phenotype? This question is of importance to NASA since astronauts are of an age where they likely go into missions with precursor lesions. It is important to understand whether their occupational radiation exposures lead to increased cancer risk. Tasks: 1. Evaluate normal and cancer cell lines from lung and colon for responses to low dose high LET radiation exposure 2. Establish epigenetic profiles for in vivo radiation induced cancers in mouse models of human lung and colon cancers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Reducing the uncertainties associated with cancer risk models for radiation exposures involved in space flight is of considerable importance in light of long-term occupation of the International Space Station (ISS) and other future manned missions by astronauts. A greater understanding of these carcinogenic mechanisms also benefits Earth-bound populations. Colon cancer is the third most common cancer. Precancerous lesions in the GI tract are found in approximately 1 out of every 10 people by age 40 and this incidence increases significantly with age. In addition to being the largest target for radiation effects, lung cancer is the most prevalent form of cancer in men and women. Each of these cancers has hallmark mutations associated with them. On top of these genetic changes, our understanding of carcinogenesis is complicated by epigenetic alterations to the genome. These epigenetic changes result in misregulation of gene expression for pathways such as cell cycle regulation, DNA repair, apoptosis, and extracellular matrix remodeling. A better understanding of these epigenetic mechanisms may lead to improvements in clinical treatments for lung, colon, and other cancers.
Task Progress:	The title of this grant is 'High LET Radiation Induced Carcinogenesis: Epigenetic Mechanisms in Mouse Models of Human Cancers'. The goal of the project is to provide epigenetic signatures that predict cancer progression in well-characterized mouse models of human lung and colon cancer, and also provide mechanistic in vitro studies of epigenetic radiation responses and cancer progression using normal, cancer and metastatic cancer cell lines. In 2013 I moved from the University of Maryland, Baltimore to the University of California, Irvine. When the funding was relinquished back to NASA from UMB, a final report was submitted for what was then known as NNX12AE79G. When my funding was released from NASA to the UCI, the grant was issued a new identifier, NNX13AK69G. It was also issued a new start of 8/01/2013 and end date of 7/31/2015. Additionally, because work from a separate NASA project (NNX13AK70G) demonstrated a lack of radiation responsiveness for two long interspersed nucleotide elements (LINEs) in high LET irradiated mice, we have now developed an additional panel of pyrosequencing assays for intracisternal A particles (IAP) that have shown changes in DNA methylation following exposure to chemicals. These IAP are a different class of mouse retrotransposable elements than the LINE. They are members of an endogenous retrovirus family that are also kept inactivated via high levels of DNA methylation, though. As originally written, the aims of the project were as follows: Aim 1 will use normal, cancer, and metastatic cancer cell lines to test the hypotheses that functional differences in profiles of DNA methylation and miRNA (miR) mediated gene regulation are A) associated with tumor aggressiveness, B) altered following radiation exposure to affect epigenetic carcinogenic progression, and C) differ based on the linear energy transfer (LET) of the radiation used. Aim 2 will test three hypotheses in mouse models of human lung and colon cancer. These hypotheses are A) miR expression affects carcinogenic initiation and progression by altering target gene expression, B) aberrations in gene promoter DNA methylation affect changes in gene expression contributing to the initiation/perpetuation of radiation carcinogenesis, and C) hypomethylation of repeat elements allows reactivation of the elements that may increase the likelihood of retrotransposition and genomic instability in radiation induced carcinogenesis. Because funding arrived at UCI late in the year, our first NSRL run was the NSRL14A campaign. During that run, we exposed lung cell lines to 600 MeV/n 56Fe ions and 350 MeV/n 28Si ions at doses of 10 and 50 cGy. Samples of these cell lines were snap frozen at BNL for early postirradiation time points. For later time points, the cells are now being grown up from frozen cell stocks here at UCI. Shortly, we will begin the comparisons of DNA methylation changes based on cellular genotype and radiation dose and quality. We are also isolating the RNAs in preparation for qRT-PCR for gene expression and analysis of miR mediated changes in gene expression. We have also recently completed and submitted an article describing experiments in which we evaluated the effect of dose, dose rate, and energy on cancer cell survival (Lasio G., Guerrero M., Goetz W., Lima F., Baulch J.E. Effect of varying dose-per-pulse and average dose rate in x-ray beam irradiation on cultured cell survival. Technology in Cancer Research and Treatment, submitted.). This manuscript is currently under review. I am an invited speaker at the 13th International Workshop on Radiation Damage to DNA. This meeting will be held in June of 2014 in Boston and my talk is titled 'Epigenetic responses to high and low LET radiations: effect of dose and radiation quality'. I am also an invited speaker at the 2014 Radiation Research Society meeting that will be held in Las Vegas in September of 2014. Work from NNX13AK69G will be presented at these conferences, and published later this year.
Bibliography Type:	Description: (Last Updated: 05/01/2017)

Articles in Peer-reviewed Journals	Thomas SN, Liao Z, Clark D, Chen Y, Samadani R, Mao L, Ann DK, Baulch JE, Shapiro P, Yang AJ. "Exosomal proteome profiling: a potential multi-marker cellular phenotyping tool to characterize hypoxia-induced radiation resistance in breast cancer." Proteomes. 2013 Sep 1;1(2):87-108. http://dx.doi.org/10.3390/proteomes1020087 ; PubMed PMID: 24860738 ; PubMed Central PMCID: PMC4029595 , Sep-2013